Topic D1+D2 HL Paper 2 Study Guide *[106 marks] -- markscheme*

A planet is in a circular orbit around a star. The speed of the planet is constant. The following data are given:

Mass of planet $= 8.0 \times 10^{24}$ kg
Mass of star $= 3.2 \times 10^{30}$ kg
Distance from the star to the planet $R = 4.4 \times 10^{10}$ m.

1a. Explain why a centripetal force is needed for the planet to be in a circular orbit. *[2 marks]*

Markscheme

«circular motion» involves a changing velocity ✓

«Tangential velocity» is «always» perpendicular to centripetal force/acceleration ✓

there must be a force/acceleration towards centre/star ✓

without a centripetal force the planet will move in a straight line ✓

1b. Calculate the value of the centripetal force. *[1 mark]*

Markscheme

$$F = \frac{(6.67\times10^{-11})(8\times10^{24})(3.2\times10^{30})}{(4.4\times10^{10})^2} = 8.8 \times 10^{23} \text{ «N» } \checkmark$$

A spacecraft is to be launched from the surface of the planet to escape from the star system. The radius of the planet is 9.1×10^3 km.

1c. Show that the gravitational potential due to the planet and the star at the surface of the planet is about $-5 \times 10^9\,\mathrm{J\,kg^{-1}}$. *[3 marks]*

Markscheme

V_{planet} = «−»$\frac{(6.67\times10^{-11})(8\times10^{24})}{9.1\times10^{6}}$ =«−» 5.9 × 10^7 «J kg^{-1}» ✓

V_{star} = «−»$\frac{(6.67\times10^{-11})(3.2\times10^{30})}{4.4\times10^{10}}$ =«−» 4.9 × 10^9 «J kg^{-1}» ✓

V_{planet} + V_{star} = «−» 4.9 «09» × 10^9 «J kg^{-1}» ✓

Must see substitutions and not just equations.

1d. Estimate the escape speed of the spacecraft from the planet–star system. *[2 marks]*

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Markscheme

use of $v_{\text{esc}} = \sqrt{2V}$ ✓

$v =$ 9.91 × 10^4 «m s^{-1}» ✓

The table gives data for Jupiter and three of its moons, including the radius *r* of each object.

Object	Mass/kg	*r* / m	Orbital radius around Jupiter/m
Jupiter	1.9×10^{27}	7.1×10^{7}	
Io	8.9×10^{22}	1.8×10^{6}	4.9×10^{8}
Ganymede	1.5×10^{23}	2.6×10^{6}	1.06×10^{9}
Callisto	1.1×10^{23}	2.4×10^{6}	1.88×10^{9}

2a. Calculate, for the surface of Io, the gravitational field strength g_{Io} due to the mass of Io. State an appropriate unit for your answer. *[2 marks]*

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Markscheme

« $\frac{GM}{r^2} = \frac{6.67 \times 10^{-11} \times 8.9 \times 10^{22}}{(1.8 \times 10^{6})^{2}} =$ » 1.8 ✓

$\mathrm{N\,kg^{-1}}$ ***OR*** $\mathrm{m\,s^{-2}}$ ✓

A spacecraft is to be sent from Io to infinity.

2b. Show that the $\frac{\text{gravitational potential due to Jupiter at the orbit of Io}}{\text{gravitational potential due to Io at the surface of Io}}$ is about 80. *[2 marks]*

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Markscheme

$\frac{1.9\times10^{27}}{4.9\times10^{8}}$ ***AND*** $\frac{8.9\times10^{22}}{1.8\times10^{6}}$ seen ✓

«$\frac{1.9\times10^{27}\times1.8\times10^{6}}{4.9\times10^{8}\times8.9\times10^{22}} =$ »78 ✓

*For **MP1**, potentials can be seen individually or as a ratio.*

2c. Outline, using (b)(i), why it is not correct to use the equation $\sqrt{\frac{2G\times\text{mass of Io}}{\text{radius of Io}}}$ to calculate the speed required for the spacecraft to reach infinity from the surface of Io. *[1 mark]*

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Markscheme

«this is the escape speed for Io alone but» gravitational potential / field of Jupiter must be taken into account ✓

OWTTE

2d. An engineer needs to move a space probe of mass 3600 kg from Ganymede to Callisto. Calculate the energy required to move the probe from the orbital radius of Ganymede to the orbital radius of Callisto. Ignore the mass of the moons in your calculation. *[2 marks]*

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Markscheme

$-GM_{\text{Jupiter}}\left(\frac{1}{1.88\times10^9}-\frac{1}{1.06\times10^9}\right)=$ «$5.21\times10^7\ \mathrm{J\,kg^{-1}}$» ✓

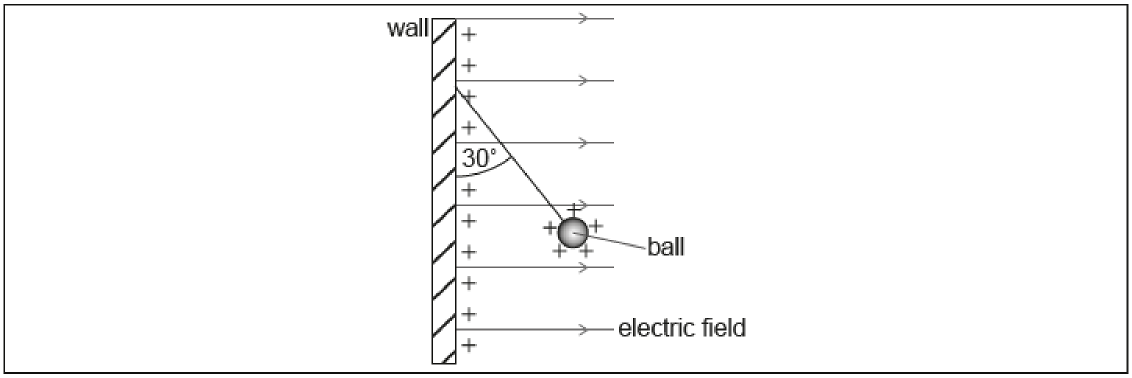
« multiplies by 3600 kg to get » 1.9×10^{11} «J» ✓

*Award **[2]** marks if factor of ½ used, taking into account orbital kinetic energies, leading to a final answer of 9.4 x 10^{10} «J».*

*Allow **ECF** from **MP1***

*Award **[2] marks** for a bald correct answer.*

A vertical wall carries a uniform positive charge on its surface. This produces a uniform horizontal electric field perpendicular to the wall. A small, positively-charged ball is suspended in equilibrium from the vertical wall by a thread of negligible mass.



3a. The charge per unit area on the surface of the wall is σ. It can be shown that the electric field strength E due to the charge on the wall is given by the equation *[2 marks]*

$E = \frac{\sigma}{2\varepsilon_0}$.

Demonstrate that the units of the quantities in this equation are consistent.

Markscheme

identifies units of σ as $\mathrm{C\,m^{-2}}$ ✓

$\frac{C}{m^2} \times \frac{Nm^2}{C^2}$ seen and reduced to $\mathrm{N\,C^{-1}}$ ✓

Accept any analysis (eg dimensional) that yields answer correctly

3b. The thread makes an angle of 30° with the vertical wall. The ball has a mass of 0.025 kg. *[3 marks]*

Determine the horizontal force that acts on the ball.

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Markscheme

horizontal force F on ball $= T \sin 30$ ✓

$T = \frac{mg}{\cos 30}$ ✓

F« $= mg \tan 30 = 0.025 \times 9.8 \times \tan 30$» $= 0.14$«N» ✓

Allow g = 10 N kg^{-1}

*Award **[3] marks** for a bald correct answer.*

*Award **[1max]** for an answer of zero, interpreting that the horizontal force refers to the horizontal component of the net force.*

3c. The charge on the ball is 1.2×10^{-6} C. Determine σ. *[2 marks]*

Markscheme

$E = \frac{0.14}{1.2 \times 10^{-6}}$ « $= 1.2 \times 10^{5}$ » ✓

$\sigma =$ « $\frac{2 \times 8.85 \times 10^{-12} \times 0.14}{1.2 \times 10^{-6}}$ » $= 2.1 \times 10^{-6}$ « $\mathrm{C\,m^{-2}}$ » ✓

*Allow **ECF** from the calculated F in (b)(i)*

*Award **[2]** for a bald correct answer.*

3d. The thread breaks. Explain the initial subsequent motion of the ball. *[3 marks]*

Markscheme

horizontal/repulsive force and vertical force/pull of gravity act on the ball ✓

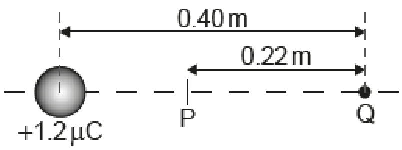
so ball has constant acceleration/constant net force ✓

motion is in a straight line ✓

at 30° to vertical away from wall/along original line of thread ✓

The centre of the ball, still carrying a charge of 1.2×10^{-6} C, is now placed 0.40 m from a point charge Q. The charge on the ball acts as a point charge at the centre of the ball.

P is the point on the line joining the charges where the electric field strength is zero. The distance PQ is 0.22 m.



3e. Calculate the charge on Q. State your answer to an appropriate number of significant figures. *[3 marks]*

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Markscheme

$\frac{Q}{0.22^2} = \frac{1.2\times10^{-6}}{0.18^2}$ ✓

$«+»1.8 \times 10^{-6}$ «C»✓

2sf ✓

*Do not award **MP2** if charge is negative*

*Any answer given to 2 sig figs scores **MP3***

3f. Outline, without calculation, whether or not the electric potential at P is zero. *[2 marks]*

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Markscheme

work must be done to move a «positive» charge from infinity to P «as both charges are positive»
OR
reference to both potentials positive and added
OR
identifies field as gradient of potential and with zero value ✓

therefore, point P is at a positive / non-zero potential ✓

*Award **[0]** for bald answer that P has non-zero potential*

The moon Phobos moves around the planet Mars in a circular orbit.

4a. Outline the origin of the force that acts on Phobos. *[1 mark]*

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Markscheme

gravitational attraction/force/field «of the planet/Mars» ✔

Do not accept “gravity”.

4b. Outline why this force does no work on Phobos. *[1 mark]*

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Markscheme

the force/field and the velocity/displacement are at 90° to each other ***OR***

there is no change in GPE of the moon/Phobos ✔

4c. The orbital period *T* of a moon orbiting a planet of mass *M* is given by *[3 marks]*

$$\frac{R^3}{T^2} = kM$$

where *R* is the average distance between the centre of the planet and the centre of the moon.

Show that $k = \frac{G}{4\pi^2}$

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Markscheme

ALTERNATE 1

«using fundamental equations»

use of Universal gravitational force/acceleration/orbital velocity equations ✔

equating to centripetal force or acceleration. ✔

rearranges to get $k = \frac{G}{4\pi^2}$ ✔

ALTERNATE 2

«starting with $\frac{R^3}{T^2} = kM$»

substitution of proper equation for T from orbital motion equations ✔

substitution of proper equation for M ***OR*** R from orbital motion equations ✔

rearranges to get $k = \frac{G}{4\pi^2}$ ✔

4d. The following data for the Mars–Phobos system and the Earth–Moon system are available: *[2 marks]*

Mass of Earth = 5.97×10^{24} kg

The Earth–Moon distance is 41 times the Mars–Phobos distance.

The orbital period of the Moon is 86 times the orbital period of Phobos.

Calculate, in kg, the mass of Mars.

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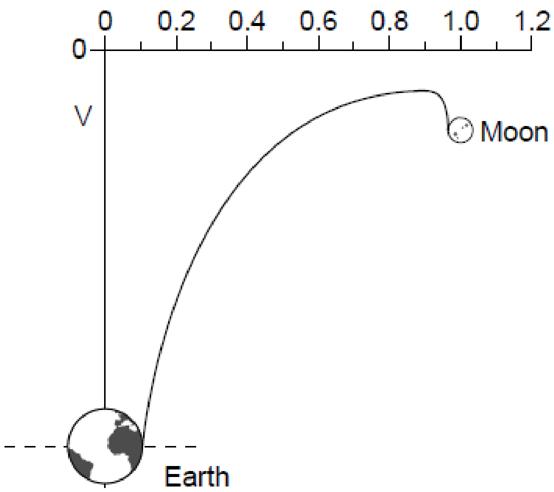
Markscheme

$m_{\text{Mars}} = \left(\frac{R_{\text{Mars}}}{R_{\text{Earth}}}\right)^3 \left(\frac{T_{\text{Earth}}}{T_{Mars}}\right)^2 m_{\text{Earth}}$ or other consistent re-arrangement ✔

6.4×10^{23} «kg» ✔

4e. The graph shows the variation of the gravitational potential between the Earth and Moon with distance from the centre of the Earth. The distance from the Earth is expressed as a fraction of the total distance between the centre of the Earth and the centre of the Moon. *[3 marks]*



Determine, using the graph, the mass of the Moon.

Markscheme

read off separation at maximum potential 0.9 ✔

equating of gravitational field strength of earth and moon at that location ***OR***

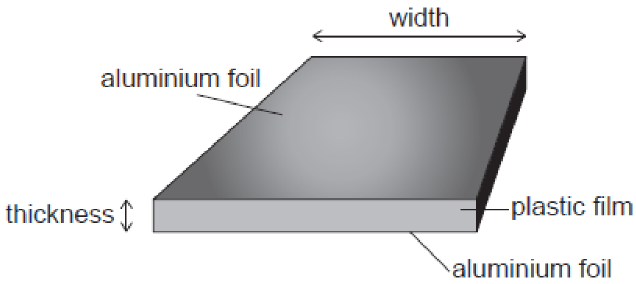
$\frac{M_E}{0.9^2} = \frac{M_M}{0.1^2}$ ✔

7.4×10^{22} «kg» ✔

Allow ECF from MP1

A student makes a parallel-plate capacitor of capacitance 68 nF from aluminium foil and plastic film by inserting one sheet of plastic film between two sheets of aluminium foil.

diagram not to scale



The aluminium foil and the plastic film are 450 mm wide.

The plastic film has a thickness of 55 μm and a permittivity of 2.5×10^{-11} C^2 N^{-1} m^{-2}.

5a. Calculate the total length of aluminium foil that the student will require. *[3 marks]*

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Markscheme

length = $\frac{d \times C}{\text{width} \times \varepsilon}$ ✔

= 0.33 «m» ✔

so 0.66/0.67 «m» «as two lengths required» ✔

5b. The plastic film begins to conduct when the electric field strength in it exceeds 1.5 MN C^{-1}. Calculate the maximum charge that can be stored on the capacitor. *[2 marks]*

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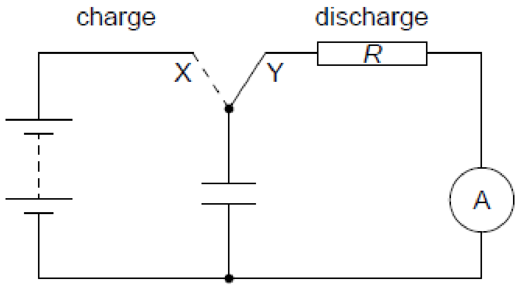
Markscheme

$1.5 \times 10^{6} \times 55 \times 10^{-6} = 83$ «V» ✔

q «= CV»$= 5.6 \times 10^{-6}$ «C»✔

The student uses a switch to charge and discharge the capacitor using the circuit shown. The ammeter is ideal.



The emf of the battery is 12 V.

5c. The resistor *R* in the circuit has a resistance of 1.2 kΩ. Calculate the time taken for the charge on the capacitor to fall to 50 % of its fully charged value. *[3 marks]*

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Markscheme

$0.5 = e^{-\frac{t}{RC}} = e^{-\frac{t}{1200 \times 6.8 \times 10^{-8}}}$

t = «−» 1200 × 6.8 × 10^{-8} × ln0.5 ✔

5.7 × 10^{-5} «s» ✔

OR

use of $t_{\frac{1}{2}} = RC \times \ln 2$ ✔

1200 × 6.8 × 10^{-8} × 0.693 ✔

5.7 × 10^{-5} «s» ✔

5d. The ammeter is replaced by a coil. Explain why there will be an induced emf in the coil while the capacitor is discharging. *[2 marks]*

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Markscheme

mention of Faraday’s law ✔

indicating that changing current in discharge circuit leads to change in flux in coil/change in magnetic field «and induced emf» ✔

5e. Suggest **one** change to the discharge circuit, apart from changes to the coil, that will increase the maximum induced emf in the coil. *[2 marks]*

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Markscheme

decrease/reduce ✔

resistance (R) ***OR*** capacitance (C) ✔

A planet of mass *m* is in a circular orbit around a star. The gravitational potential due to the star at the position of the planet is *V*.

6a. Show that the total energy of the planet is given by the equation shown. *[2 marks]*

$E = \frac{1}{2}mV$

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Markscheme

$E = \frac{1}{2}m\frac{GM}{r} - \frac{GMm}{r} = -\frac{1}{2}\frac{GMm}{r}$ ✔

comparison with $V = -\frac{GM}{r}$ ✔

«to give answer»

6b. Suppose the star could contract to half its original radius without any loss of mass. Discuss the effect, if any, this has on the total energy of the planet. *[2 marks]*

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Markscheme

ALTERNATIVE 1

«at the position of the planet» the potential depends only on the mass of the star /does not depend on the radius of the star ✔

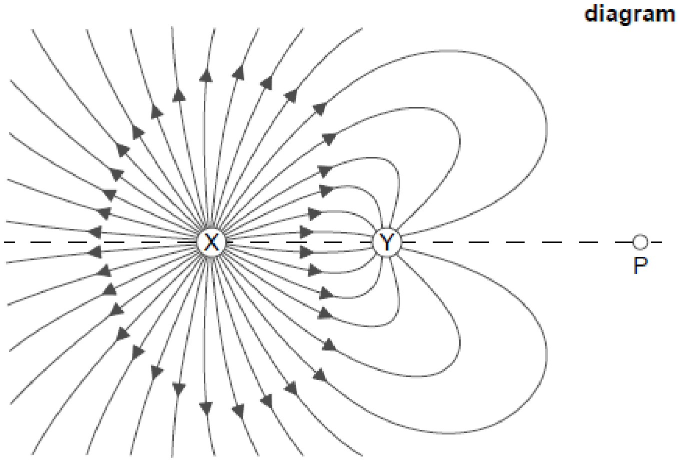
the potential will not change and so the energy will not change ✔

ALTERNATIVE 2

r / distance between the centres of the objects / orbital radius remains unchanged ✔

since $E_{Total} = -\frac{1}{2}\frac{GMm}{r}$, energy will not change ✔

6c. The diagram shows some of the electric field lines for two fixed, charged particles X and Y. *[2 marks]*



The magnitude of the charge on X is Q and that on Y is q. The distance between X and Y is 0.600 m. The distance between P and Y is 0.820 m.

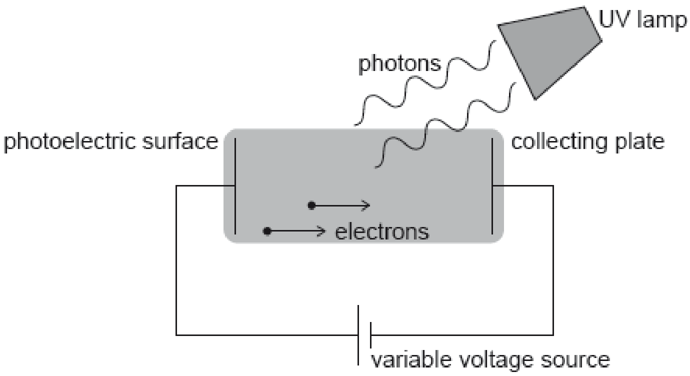
At P the electric field is zero. Determine, to **one** significant figure, the ratio $\frac{Q}{q}$.

Markscheme

$$\frac{kQ}{(0.600+0.820)^2} = \frac{kq}{0.820^2} \checkmark$$

$$\frac{Q}{q} = « \frac{(0.600+0.820)^2}{0.820^2} = 2.9988 \approx » 3 \checkmark$$

Hydrogen atoms in an ultraviolet (UV) lamp make transitions from the first excited state to the ground state. Photons are emitted and are incident on a photoelectric surface as shown.



7a. Show that the energy of photons from the UV lamp is about 10 eV. *[2 marks]*

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Markscheme

E_1 = –13.6 «eV» $E_2 = -\frac{13.6}{4}$ = –3.4 «eV»

energy of photon is difference $E_2 - E_1$ = 10.2 «≈ 10 eV»

Must see at least 10.2 eV.

[2 marks]

The photons cause the emission of electrons from the photoelectric surface. The work function of the photoelectric surface is 5.1 eV.

7b. Calculate, in J, the maximum kinetic energy of the emitted electrons. *[2 marks]*

Markscheme

10 – 5.1 = 4.9 «eV»

$4.9 \times 1.6 \times 10^{-19} = 7.8 \times 10^{-19}$ «J»

Allow 5.1 *if* 10.2 *is used to give* 8.2×10^{-19} «J».

7c. Suggest, with reference to conservation of energy, how the variable voltage source can be used to stop all emitted electrons from reaching the collecting plate. *[2 marks]*

Markscheme

EPE produced by battery

exceeds maximum KE of electrons / electrons don’t have enough KE

For first mark, accept explanation in terms of electric potential energy difference of electrons between surface and plate.

[2 marks]

7d. The variable voltage can be adjusted so that no electrons reach the collecting plate. Write down the minimum value of the voltage for which no electrons reach the collecting plate. *[1 mark]*

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Markscheme

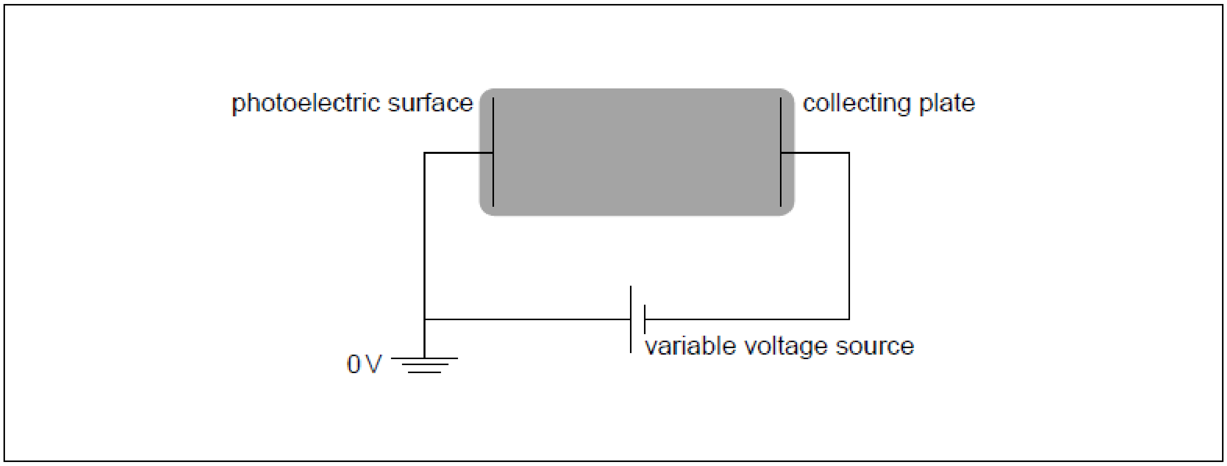
4.9 «V»

Allow 5.1 if 10.2 is used in (b)(i).

Ignore sign on answer.

[1 mark]

The electric potential of the photoelectric surface is 0 V. The variable voltage is adjusted so that the collecting plate is at –1.2 V.

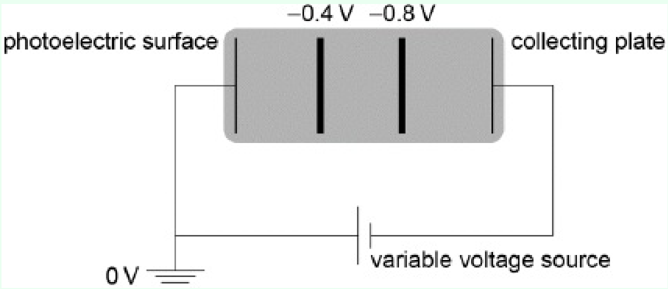


7e. On the diagram, draw and label the equipotential lines at –0.4 V and –0.8 V. *[2 marks]*

Markscheme

two equally spaced vertical lines (judge by eye) at approximately 1/3 and 2/3

labelled correctly



[2 marks]

7f. An electron is emitted from the photoelectric surface with kinetic energy 2.1 eV. Calculate the speed of the electron at the collecting plate. *[2 marks]*

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Markscheme

kinetic energy at collecting plate = 0.9 «eV»

speed = «$\sqrt{\frac{2\times0.9\times1.6\times10^{-19}}{9.11\times10^{-31}}}$» = 5.6×10^{5} «ms^{-1}»

Allow ECF from MP1

[2 marks]

A planet has radius *R*. At a distance *h* above the surface of the planet the gravitational field strength is *g* and the gravitational potential is *V*.

8a. State what is meant by gravitational field strength. *[1 mark]*

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Markscheme

the «gravitational» force per unit mass exerted on a point/small/test mass

[1 mark]

8b. Show that $V = -g(R + h)$. *[2 marks]*

Markscheme

at height h potential is $V = -\frac{GM}{(R+h)}$

field is $g = \frac{GM}{(R+h)^2}$

«dividing gives answer»

Do not allow an answer that starts with $g = -\frac{\Delta V}{\Delta r}$ *and then cancels the deltas and substitutes* $R + h$

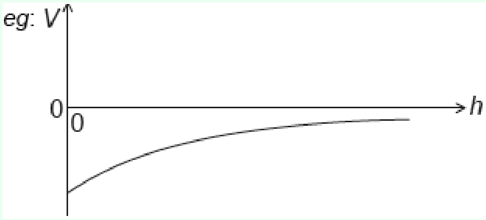
[2 marks]

8c. Draw a graph, on the axes, to show the variation of the gravitational potential V of the planet with height h above the surface of the planet. *[2 marks]*

Markscheme

correct shape and sign

non-zero negative vertical intercept



[2 marks]

8d. A planet has a radius of 3.1×10^6 m. At a point P a distance 2.4×10^7 m above the surface of the planet the gravitational field strength is 2.2 N kg^{-1}. Calculate the gravitational potential at point P, include an appropriate unit for your answer. *[1 mark]*

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Markscheme

V = «–2.2 × ($3.1 \times 10^6 + 2.4 \times 10^7$) =» «–» 6.0×10^7 J kg^{-1}

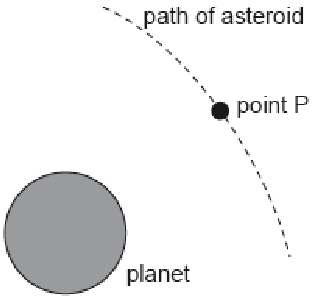
Unit is essential

Allow eg MJ kg^{-1} if power of 10 is correct

Allow other correct SI units eg m^{2}s^{-2}, N m kg^{-1}

[1 mark]

8e. The diagram shows the path of an asteroid as it moves past the planet. *[3 marks]*



When the asteroid was far away from the planet it had negligible speed. Estimate the speed of the asteroid at point P as defined in (b).

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Markscheme

total energy at P = 0 / KE gained = GPE lost

«$\frac{1}{2}mv^2 + mV = 0 \Rightarrow$» $v = \sqrt{-2V}$

$v =$ «$\sqrt{2 \times 6.0 \times 10^7}$ =» 1.1×10^4 «ms^{-1}»

*Award **[3]** for a bald correct answer*

Ignore negative sign errors in the workings

Allow ECF from 6(b)

[3 marks]

8f. The mass of the asteroid is 6.2×10^{12} kg. Calculate the gravitational force experienced by the **planet** when the asteroid is at point P. *[2 marks]*

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Markscheme

ALTERNATIVE 1

force on asteroid is «$6.2 \times 10^{12} \times 2.2$ =» 1.4×10^{13} «N»

«by Newton's third law» this is also the force on the planet

ALTERNATIVE 2

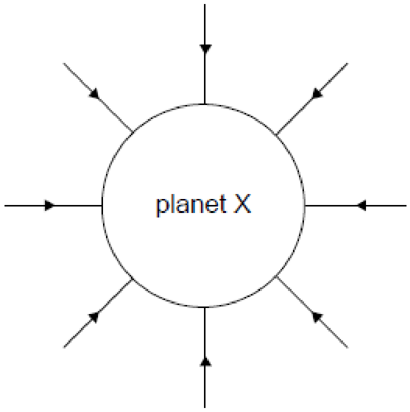
mass of planet = 2.4×10^{25} «kg» «from $V = -\frac{GM}{(R+h)}$»

force on planet «$\frac{GMm}{(R+h)^2}$» = 1.4×10^{13} «N»

MP2 must be explicit

[2 marks]

The diagram shows the gravitational field lines of planet X.



9a. Outline how this diagram shows that the gravitational field strength of planet X decreases with distance from the surface. *[1 mark]*

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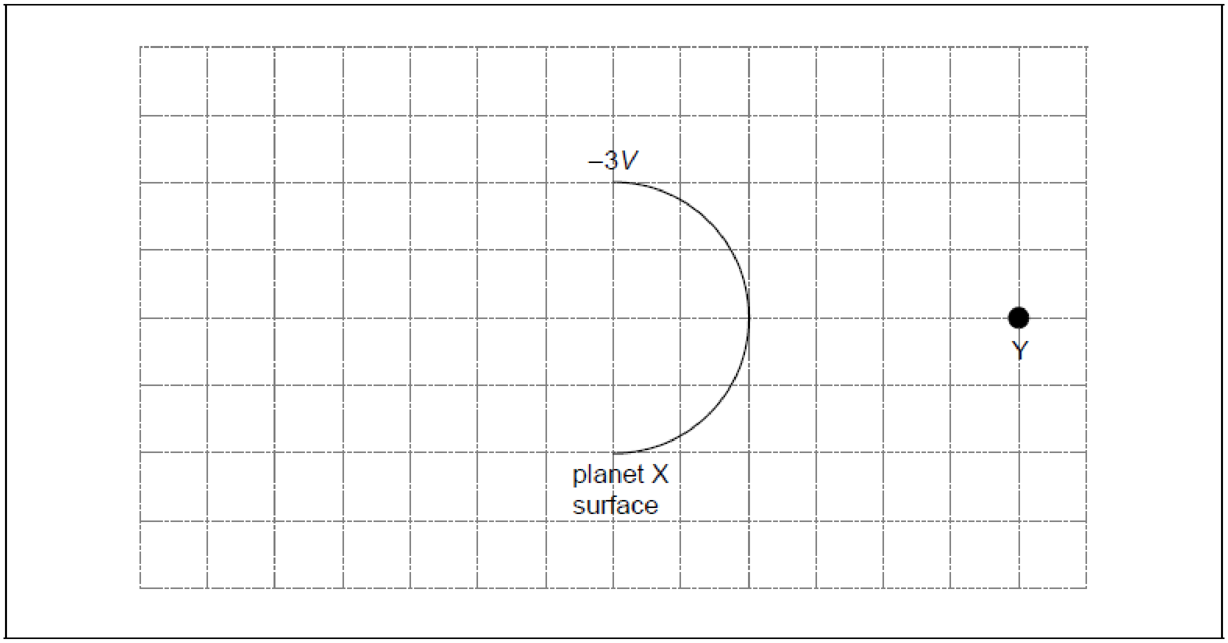
Markscheme

the field lines/arrows are further apart at greater distances from the surface

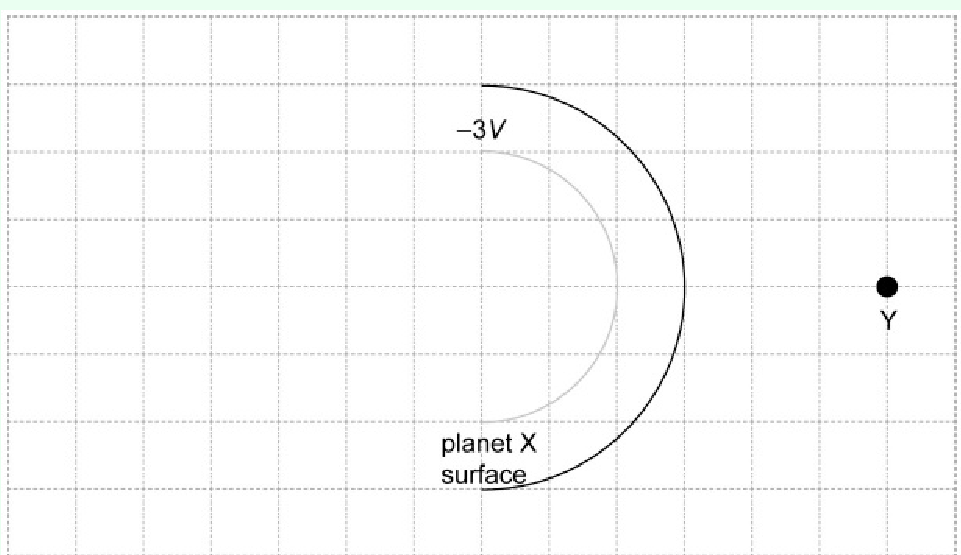
9b. The diagram shows part of the surface of planet X. The gravitational potential at the surface of planet X is $-3V$ and the gravitational potential at point Y is $-V$. *[2 marks]*



Sketch on the grid the equipotential surface corresponding to a gravitational potential of $-2V$.

Markscheme

circle centred on Planet X
three units from Planet X centre



9c. A meteorite, very far from planet X begins to fall to the surface with a negligibly small initial speed. The mass of planet X is 3.1×10^{21} kg and its radius is 1.2×10^{6} m. The planet has no atmosphere. Calculate the speed at which the meteorite will hit the surface. *[3 marks]*

Markscheme

loss in gravitational potential = $\frac{6.67\times10^{-11}\times3.1\times10^{21}}{1.2\times10^{6}}$

«= 1.72×10^{5} JKg^{-1}»

equate to $\frac{1}{2}v^2$

v = 590 «m s^{-1}»

Allow ECF from MP1.

9d. At the instant of impact the meteorite which is made of ice has a temperature of 0 °C. Assume that all the kinetic energy at impact gets transferred into internal energy in the meteorite. Calculate the percentage of the meteorite's mass that melts. The specific latent heat of fusion of ice is 3.3×10^{5} J kg^{-1}. *[2 marks]*

Markscheme

available energy to melt one kg 1.72×10^5 «J»

fraction that melts is $\frac{1.72 \times 10^5}{3.3 \times 10^5}$ = 0.52 ***OR*** 52%

Allow ECF from MP1.

Allow 53% from use of 590 ms^{-1}.

The gravitational potential due to the Sun at its surface is -1.9×10^{11} J kg^{-1}. The following data are available.

Mass of Earth	$= 6.0 \times 10^{24}$ kg
Distance from Earth to Sun	$= 1.5 \times 10^{11}$ m
Radius of Sun	$= 7.0 \times 10^{8}$ m

10a. Outline why the gravitational potential is negative. *[2 marks]*

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Markscheme

potential is defined to be zero at infinity

so a positive amount of work needs to be supplied for a mass to reach infinity

10b. The gravitational potential due to the Sun at a distance *r* from its centre is V_S. Show that *[1 mark]*

rV_S = constant.

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Markscheme

$V_S = -\frac{GM}{r}$ so $r \times V_S$ «= $-GM$» = constant because *G* and *M* are constants

10c. Calculate the gravitational potential energy of the Earth in its orbit around the Sun. Give your answer to an appropriate number of significant figures. *[2 marks]*

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Markscheme

$GM = 1.33 \times 10^{20}$ «J m kg^{-1}»

GPE at Earth orbit «= $-\frac{1.33\times10^{20}\times6.0\times10^{24}}{1.5\times10^{11}}$» = «–» 5.3×10^{33} «J»

*Award **[1 max]** unless answer is to 2 sf.*

Ignore addition of Sun radius to radius of Earth orbit.

10d. Calculate the total energy of the Earth in its orbit. *[2 marks]*

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Markscheme

ALTERNATIVE 1

work leading to statement that kinetic energy $\frac{GMm}{2r}$, ***AND*** kinetic energy evaluated to be «+» 2.7 x 10^{33} «J»

energy «= PE + KE = answer to (b)(ii) + 2.7 x 10^{33}» = «–» 2.7 x 10^{33} «J»

ALTERNATIVE 2

statement that kinetic energy is $= -\frac{1}{2}$ gravitational potential energy in orbit

so energy «$= \frac{\text{answer to (b)(ii)}}{2}$» = «–» 2.7 x 10^{33} «J»

Various approaches possible.

10e. An asteroid strikes the Earth and causes the orbital speed of the Earth to suddenly decrease. Suggest the ways in which the orbit of the Earth will change. *[2 marks]*

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Markscheme

«KE will initially decrease so» total energy decreases
OR
«KE will initially decrease so» total energy becomes more negative

Earth moves closer to Sun

new orbit with greater speed «but lower total energy»

changes ellipticity of orbit

10f. Outline, in terms of the force acting on it, why the Earth remains in a circular orbit around the Sun. *[2 marks]*

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Markscheme

centripetal force is required

and is provided by gravitational force between Earth and Sun

*Award **[1 max]** for statement that there is a "centripetal force of gravity" without further qualification.*

11a. Outline what is meant by escape speed. *[1 mark]*

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Markscheme

speed to reach infinity/zero gravitational field
OR
speed to escape gravitational pull/effect of planet’s gravity

Do not allow reference to leaving/escaping an orbit.
Do not allow “escaping the atmosphere”.

11b. A probe is launched vertically upwards from the surface of a planet with a speed *[3 marks]*

$$v = \frac{3}{4} v_{\mathrm{esc}}$$

where v_{esc} is the escape speed from the planet. The planet has no atmosphere.

Determine, in terms of the radius of the planet *R*, the maximum height from the surface of the planet reached by the probe.

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Markscheme

«kinetic energy at take off» $= \frac{9}{16} \times \frac{GMm}{R}$

kinetic energy at take off + «gravitational» potential energy = «gravitational» potential energy at maximum height

OR
$\frac{9}{16} \times \frac{GMm}{R} - \frac{GMm}{R} = -\frac{GMm}{r}$

solves for *r* and subtracts *R* from answer $= \frac{9R}{7}$

*Award **[0]** for work that assumes constant g.*

11c. The total energy of a probe in orbit around a planet of mass M is *[3 marks]*

$E = -\frac{GMm}{2r}$ where m is the mass of the probe and r is the orbit radius.

A probe in low orbit experiences a small frictional force. Suggest the effect of this force on the speed of the probe.

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Markscheme

energy reduces/lost

radius decreases

speed increases

Do not allow “kinetic energy reduces” for MP1

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